

Bernard Benson, PhD

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Research Interests

Machine learning and deep learning for the physical sciences; heliophysics and space-weather forecasting; time-series modeling and uncertainty quantification; natural language processing and large language models (retrieval-augmented generation, agentic and MCP-based tools); computer vision.

Education

Ph.D., Electrical and Computer Engineering 2016 – 2021

University of Alabama in Huntsville, Huntsville, AL

M.S., Electrical and Computer Engineering 2011 – 2012

University of Alabama in Huntsville, Huntsville, AL

B.Tech., Electrical and Electronics Engineering 2006 – 2010

Geethanjali College of Engineering and Technology

Appointments & Professional Experience

Research Scientist IV May 2025 – Present

NASA ODSI, University of Alabama in Huntsville, Huntsville, AL

- Developed ensemble large language models for multi-label text classification of scientific documents.
- Built a search wrapper modernizing the backend of NASA's Science Discovery Engine (SDE) using OpenSearch.
- Researched and developed MCP tools for PDS4 and SDE data-search agents.

Generative AI Solutions Architect Dec 2024 – May 2025

Megan Soft VA, Remote

- Developed an evaluation framework for generative-AI chatbots and RAG systems using DeepEval and Ragas.
- Established KPIs and testing procedures to optimize performance and user satisfaction.
- Designed multi-chatbot orchestration strategies for agents with overlapping capabilities.
- Designed and implemented recommender systems for pharmacy benefits management.

Data Scientist Aug 2021 – Nov 2024

McLeod Software, Birmingham, AL

- Led development and deployment of ML models predicting freight rates for over 1,100 trucking organizations, improving pricing accuracy and efficiency.
- Designed and deployed retrieval-augmented-generation (RAG) chatbots for customer support, reducing resolution times.
- Built and deployed LayoutLMv2-based models to automate document classification.
- Led research and analysis of AI applications across the transportation and logistics industry.
- Represented McLeod Software as a conference speaker at industry events on AI in transportation and logistics.

Applied ML Researcher Jun 2021 – Aug 2021

Frontier Development Lab (NASA / SETI Institute), Remote

- Applied machine learning to model the effect of solar drag on satellite orbits as part of a NASA-sponsored research project.

- Developed a multivariate time-series model based on deep neural network ensembles to forecast space-weather indices.
- Earned a team excellence award for the project.

Graduate Teaching Assistant

Aug 2016 – May 2021

University of Alabama in Huntsville, Huntsville, AL

- Taught and supervised lab sections for CPE211L Introduction to Programming (C++).
- Taught EE316 Electrical Circuits and Electrical Design Lab (Summer 2017).

RF Engineer

Mar 2013 – Mar 2016

Nexgen Wireless / Global Telecom Associates, Milwaukee, WI

- Analyzed drive-test data and network logs to identify and resolve coverage gaps and interference issues, improving network performance.
- Visualized network performance with heatmaps and coverage plots to support data-driven optimization.

Publications

Refereed Journal Articles

- [1] **B. Benson**, W. D. Pan, A. Prasad, G. A. Gary, and Q. Hu, “Forecasting Solar Cycle 25 Using Deep Neural Networks,” *Solar Physics*, vol. 295, no. 5, art. 65, 2020.
- [2] T. Singh, **B. Benson**, S. A. Z. Raza, T. K. Kim, N. V. Pogorelov, W. P. Smith, and C. N. Arge, “Improving the Arrival-Time Estimates of Coronal Mass Ejections by Using Magnetohydrodynamic Ensemble Modeling, Heliospheric Imager Data, and Machine Learning,” *The Astrophysical Journal*, vol. 948, no. 2, art. 78, 2023.
- [3] **B. Benson**, W. D. Pan, G. A. Gary, Q. Hu, and T. Staudinger, “Determining the Parameter for the Linear Force-Free Magnetic Field Model with Multi-Dipolar Configurations Using Deep Neural Networks,” *Astronomy and Computing*, vol. 26, pp. 50–60, 2019.
- [4] T. S. Newman, C. W. Hall, L. Farris, T. Singh, N. Pogorelov, **B. Benson**, S. Raza, *et al.*, “Solar Flare Forecasting Using Machine Learning and SDO/HMI Data: A Multiple Machine Learning Model and Data Curation Technique Comparison Study,” *The Astrophysical Journal Supplement Series*, vol. 280, no. 2, art. 54, 2025.
- [5] **B. Benson**, W. D. Pan, A. Prasad, G. A. Gary, and Q. Hu, “On the Estimation of the SHARP Parameter MEANALP from AIA Images Using Deep Neural Networks,” *Solar Physics*, vol. 296, no. 11, art. 163, 2021.

Conference and Workshop Papers, Abstracts, and Posters

- [1] S. Bonasera, G. Acciarini, J. A. Pérez-Hernández, **B. Benson**, E. Brown, *et al.*, “Dropout and Ensemble Networks for Thermospheric Density Uncertainty Estimation,” *Bayesian Deep Learning Workshop, NeurIPS 2021*, 2021.
- [2] E. J. E. Brown, S. Bonasera, **B. Benson**, J. A. Pérez-Hernández, G. Acciarini, *et al.*, “Learning the Solar Latent Space: Sigma-Variational Autoencoders for Multiple Channel Solar Imaging,” *Fourth Workshop on Machine Learning and the Physical Sciences, NeurIPS 2021*, 2021.
- [3] **B. Benson**, E. Brown, S. Bonasera, G. Acciarini, J. A. Pérez-Hernández, *et al.*, “Simultaneous Multivariate Forecast of Space Weather Indices Using Deep Neural Network Ensembles,” *Fourth Workshop on Machine Learning and the Physical Sciences, NeurIPS 2021*, 2021.
- [4] **B. Benson**, Z. Jiang, W. D. Pan, G. A. Gary, and Q. Hu, “Determination of Linear Force-Free Magnetic Field Constant Alpha Using Deep Learning,” *International Conference on Computational Science and Computational Intelligence (CSCI)*, 2017.

- [5] T. Singh, L. Farris, C. Hall, T. Newman, **B. Benson**, S. Raza, and N. Pogorelov, “Solar Flare Forecasting Using Multiple Machine Learning Models and SDO/HMI Data,” *SHINE 2024 Workshop*, 2024.
- [6] R. Hooda, W. D. Pan, and **B. Benson**, “Transform Decomposition Switching for Efficient Attribute Compression of 3D Point Clouds Using Neural Networks,” *International Conference on Computational Science and Computational Intelligence (CSCI)*, 2022.
- [7] T. Singh, **B. Benson**, *et al.*, “Improving the Arrival-Time Prediction of Coronal Mass Ejections Using Magnetohydrodynamic Ensemble Modeling, Heliospheric Imager Data, and Machine Learning,” *Proceedings of the 2nd Machine Learning in Heliophysics Conference*, 2022.
- [8] C. Davis, B. Praveen, E. Foshee, K. Bugbee, D. Sharma, N. Pantha, S. Awale, **B. Benson**, *et al.*, “LLMs, Encoders, and Agentic Workflows for Curation and Discovery in NASA’s Science Discovery Engine,” *AGU Fall Meeting 2025*, 2025.
- [9] T. Singh, C. Hall, T. Newman, **B. Benson**, S. Raza, and N. Pogorelov, “Solar Flare Forecasting Using Machine Learning and SDO/HMI Data: A Comparison of Multiple ML Models and AR Parameter Time Series,” *SHINE 2023 Workshop*, 2023.
- [10] T. Singh, **B. Benson**, *et al.*, “Forecasting Solar Flares Using the Time Evolution of Active Regions and Machine Learning Techniques,” *NASA Proposal ID 21-SWR202R_2-0011*, 2021.

Invited Talks

- “Deep Neural Networks Applied to Three Problems in Heliophysics: Coronal Magnetic Field Modeling, Free Magnetic Energy, and Forecasting Solar Activity,” NASA Marshall Space Flight Center, Heliophysics Division, Sep. 2020.
- “Applications of AI in Logistics and Supply Chain,” Auburn University, Harbert College of Business, Oct. 2023.
- “AI in the Supply Chain,” Mississippi State University, College of Business, Mar. 2024.

Teaching Experience

- **CPE211L — Introduction to Programming (C++)**, undergraduate lab instructor, University of Alabama in Huntsville.
- **EE316 — Electrical Circuits and Electrical Design Lab**, undergraduate lab instructor, University of Alabama in Huntsville (Summer 2017).

Honors & Awards

- Team Excellence Award, Frontier Development Lab (NASA / SETI Institute), 2021.

Professional Service

Reviewer

- Research proposals, NASA Heliophysics.
- *Journal of Geophysical Research: Machine Learning and Computation*.
- *Journal of Atmospheric and Solar-Terrestrial Physics*.
- *Solar Physics*.
- Journals of the American Astronomical Society.
- IEEE SoutheastCon — AI and ML tracks (Mar. 2021).

Session Chair

- IEEE SoutheastCon — Computing track (Mar. 2021).

Selected Research Projects

- **Improving CME Arrival Time with Ensemble Models and Machine Learning** — regression and neural-network models to improve coronal mass ejection arrival-time predictions.
- **Simultaneous Multivariate Forecasts of Space-Weather Indices using DNN Ensembles** — LSTM ensembles forecasting space-weather indices over a full solar rotation.
- **Solar Image Processing for Flare Detection** — spatial and frequency-domain filtering to enhance coronal loops, latent-space visualization with VAEs, and relating SHARP parameters to flaring events.
- **Time-Series Forecast of Solar Cycle 25 using Deep Neural Networks** — WaveNet and LSTM models predicting the strength of Solar Cycle 25.

Technical Skills

- **Programming Languages:** Python, SQL, C++, MATLAB.
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn; CNNs, RNNs, LSTMs, Transformers; retrieval-augmented generation (RAG) and large language models.
- **Data Analysis & Visualization:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Power BI, Tableau.
- **Cloud & DevOps:** AWS, Azure, Google Cloud Platform; Docker, Git, CI/CD pipelines, REST APIs.